# Sustainable Jute Fiber Reinforced Polymer Composites (JFRPC) for Automotive Structures

Go-Kart Body Panel Development (BUET Automobile Club)

**Abstract**

This report presents the development and evaluation of **jute fiber reinforced polymer composites (JFRPC)** as a sustainable alternative material for lightweight automotive structures. Bidirectional jute fiber mats were used as reinforcement with epoxy resin as the matrix material. Composite laminates were fabricated using a **hand lay-up** process and mechanically characterized using standard testing methodologies. The results show that increasing jute fiber loading improves tensile response, hardness, and interlaminar performance, supporting the feasibility of natural fiber composites as a low-cost and locally sourced option for go-kart body panels.

## 1 Introduction

Natural fiber reinforced polymer composites have gained increasing attention in engineering applications due to their renewability, biodegradability, and low cost. Compared to synthetic reinforcements, natural fibers offer significant environmental benefits and are particularly attractive in regions where they are locally abundant.

Jute is one of the most widely available natural fibers in Bangladesh. Its availability, cost-effectiveness, and relatively good mechanical properties make it a suitable candidate for reinforcing polymer composites. This project investigates the feasibility of **jute fiber reinforced epoxy composites** for use in go-kart body structures, as an alternative to conventional synthetic composite panels.

## 2 Materials and Fabrication

### 2.1 Materials

The composite laminates were fabricated using:

- **Reinforcement:** Bidirectional jute fiber mat (locally sourced)
- **Matrix:** Epoxy resin and hardener

## 2.2 Fabrication Method

The laminates were produced using the **hand lay-up technique**. During fabrication, resin impregnation was performed through stacked layers of bidirectional jute mat. Layer rolling and compaction were applied to improve wet-out and reduce trapped air between layers.

## 2.3 Manufacturing Considerations

Composite performance is strongly affected by:

- Fiber loading and distribution
- Fiber–matrix bonding quality
- Interlaminar integrity and void content

# 3 Mechanical Testing Standards

Mechanical characterization was conducted to evaluate suitability for go-kart body panel applications. The testing standards and equipment reported include:

- **Tensile testing:** ASTM D3039 (universal testing machine, Instron 1195)
- **Impact testing:** ASTM D256
- **Hardness testing:** Rockwell hardness test (steel ball indenter)

## 3.1 Experimental Methodology

Composite laminates were fabricated using a hand lay-up technique with bidirectional jute fiber reinforcement and epoxy resin matrix.

Tensile testing was performed in accordance with ASTM D3039 using a universal testing machine (Instron 1195). The test evaluated tensile strength and modulus behavior as a function of fiber loading.

Impact resistance was evaluated following ASTM D256 to assess damage tolerance characteristics of the laminates.

Hardness testing was conducted using a Rockwell hardness tester equipped with a steel ball indenter.

The influence of fiber loading was evaluated at five reinforcement levels: 0 wt.%, 12 wt.%, 24 wt.%, 36 wt.%, and 48 wt.%.

# 4 Results

The composite laminates were successfully fabricated and tested. The reported results show strong dependence of mechanical properties on fiber loading. In particular:

- Tensile properties increase with increasing fiber loading.
- Impact strength increases with increasing fiber loading.

- Hardness increases with increasing fiber loading.
- Flexural and interlaminar properties are strongly influenced by void content.

### 4.1 Flexural Strength and Flexural Modulus vs Fiber Loading

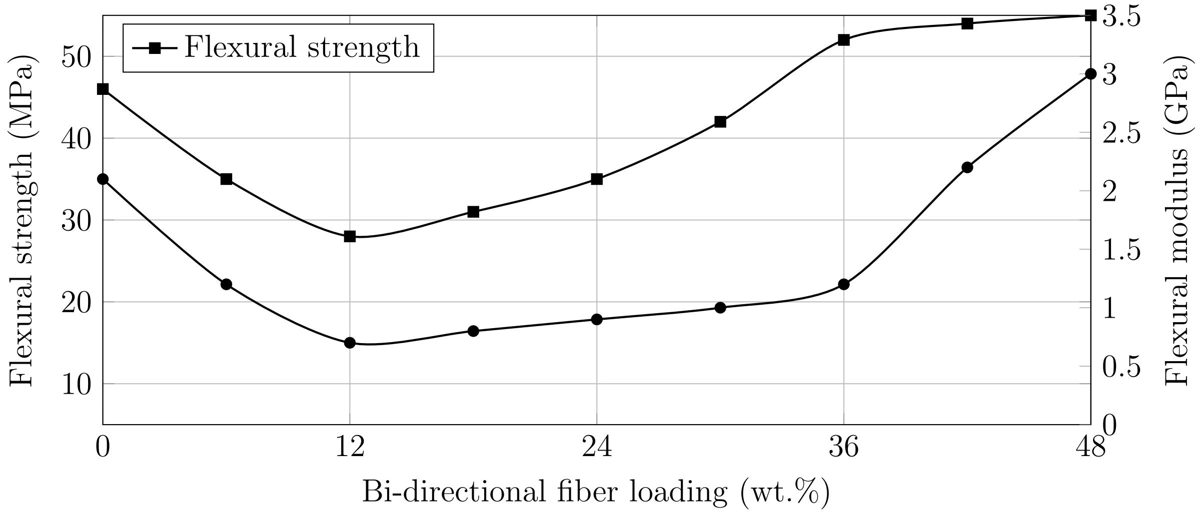


Figure 1: Flexural strength and flexural modulus as a function of bidirectional jute fiber loading.

### 4.2 Tensile Strength and Tensile Modulus vs Fiber Loading

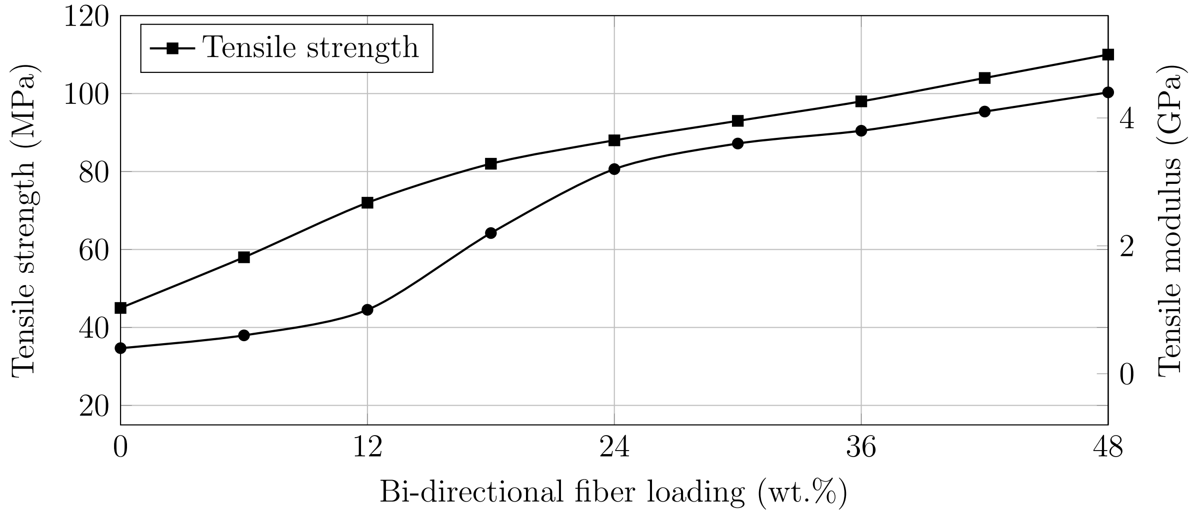


Figure 2: Tensile strength and tensile modulus as a function of bidirectional jute fiber loading.

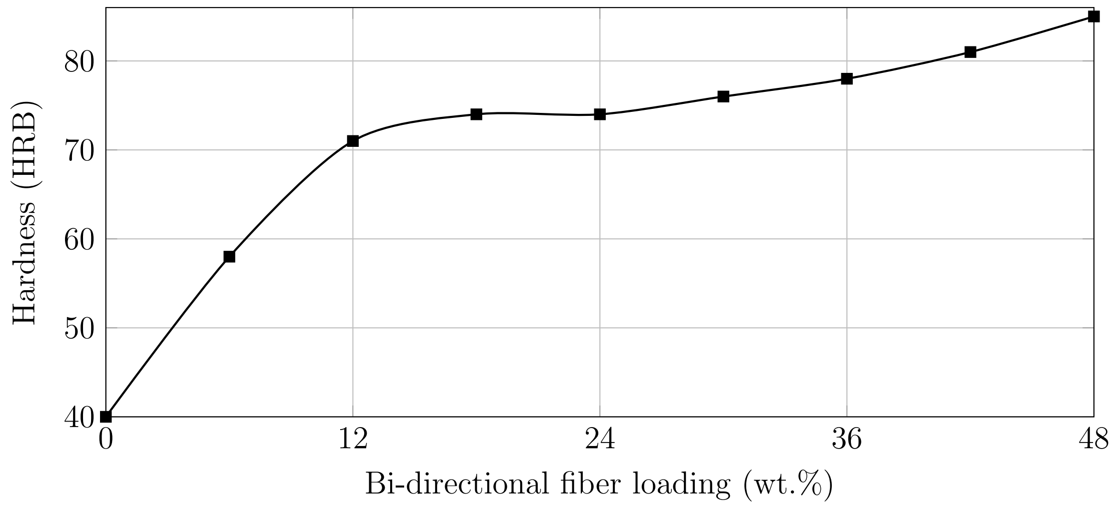


Figure 3: Hardness as a function of bidirectional jute fiber loading.

### 4.3 Hardness vs Fiber Loading

### 4.4 Interlaminar Shear Strength (ILSS) vs Fiber Loading

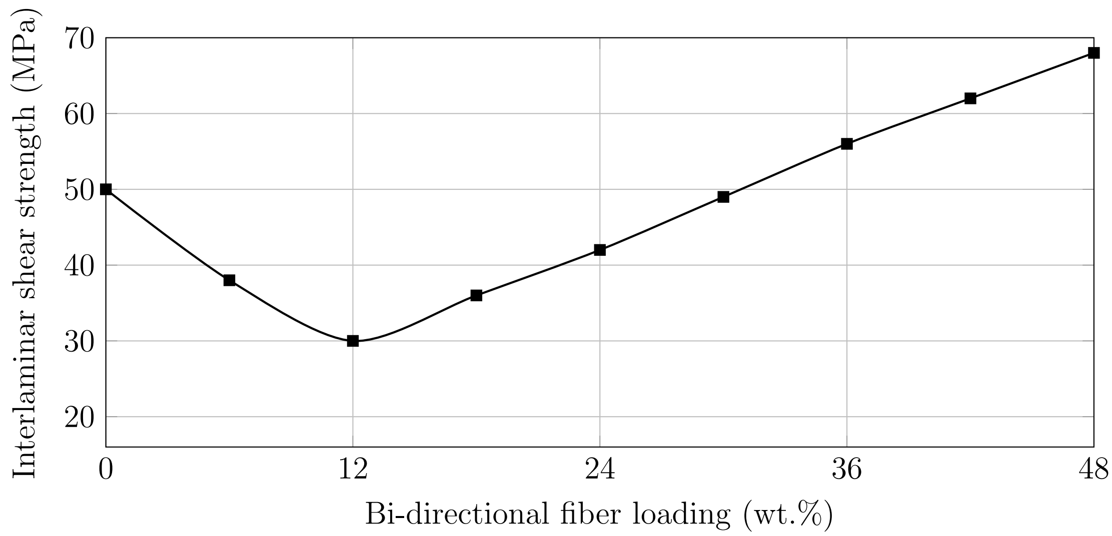


Figure 4: Interlaminar shear strength (ILSS) as a function of bidirectional jute fiber loading.

### 4.5 Results Summary Table

Table 1 summarizes the mechanical property trends as a function of bidirectional jute fiber loading.

## 5 Discussion

The results demonstrate consistent improvements in tensile performance and hardness as fiber loading increases. The interlaminar and flexural responses show greater sensitivity

Table 1: Mechanical properties versus fiber loading

| Fiber Loading (wt.%) | Flexural Strength (MPa) | Tensile Strength (MPa) | Hardness (HRB) | ILSS |
|---|---|---|---|---|
| 0 | 46 | 45 | 40 | |
| 12 | 28 | 72 | 71 | |
| 24 | 35 | 88 | 74 | |
| 36 | 52 | 98 | 78 | |
| 48 | 55 | 110 | 85 | |

to laminate integrity, consistent with the report's emphasis on void content as a dominant factor affecting flexural and interlaminar properties.

The reported results highlight that fiber loading alone does not fully determine performance; manufacturing quality (in particular void reduction and good layer compaction) is essential for achieving reliable interlaminar strength.

### 5.1 Influence of Void Content

The report identifies void content as a key parameter influencing flexural and interlaminar properties. In particular, it notes that flexural strength and interlaminar shear strength are strongly affected by trapped air between layers. Improved performance at higher fiber loading is associated with improved laminate integrity and reduced void content.

## 6 Sustainability and Material Advantage

Jute fiber reinforced polymer composites offer several environmental and economic advantages compared to conventional synthetic composites:

- **Renewability:** Jute is a biodegradable and renewable natural fiber.
- **Local availability:** Bangladesh is one of the largest producers of jute, reducing sourcing distance and cost.
- **Lower embodied energy:** Natural fibers generally require less energy for production than synthetic fibers.
- **Cost-effectiveness:** Jute fibers are economical compared to conventional synthetic reinforcements.

The use of locally sourced jute fiber not only reduces material cost but also supports rural agricultural economies. For lightweight automotive structures such as go-kart body panels, JFRPC provides an environmentally responsible alternative while maintaining adequate mechanical performance trends.

## 7 Feasibility for Go-Kart Body Panels

The mechanical performance trends support the feasibility of JFRPC for go-kart body structures such as:

- Nose cone and bumper fairings

- Side pods and aerodynamic panels
- Rear protective covers and fairings

These components benefit from low mass, adequate stiffness, impact resistance, and sustainable material sourcing.

# 8 Conclusion

Jute fiber reinforced epoxy composites were successfully fabricated using a hand lay-up technique and characterized using ASTM D3039 tensile testing, ASTM D256 impact testing, and Rockwell hardness testing. The results show that increasing bidirectional jute fiber loading improves tensile response, hardness, and interlaminar performance. The study confirms that JFRPC is a viable low-cost and sustainable alternative material for go-kart body panel applications, with laminate quality and void minimization being critical to achieving reliable performance.